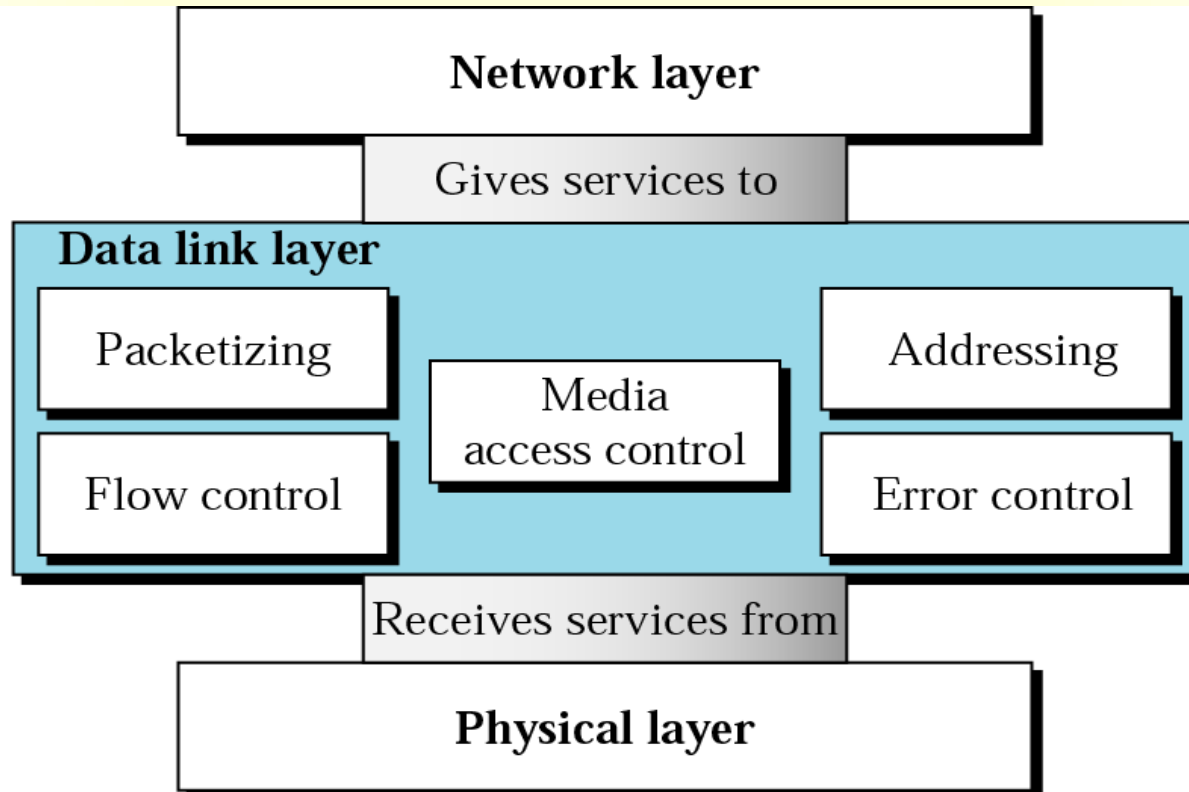


Chapter Three: The Data Link Layer

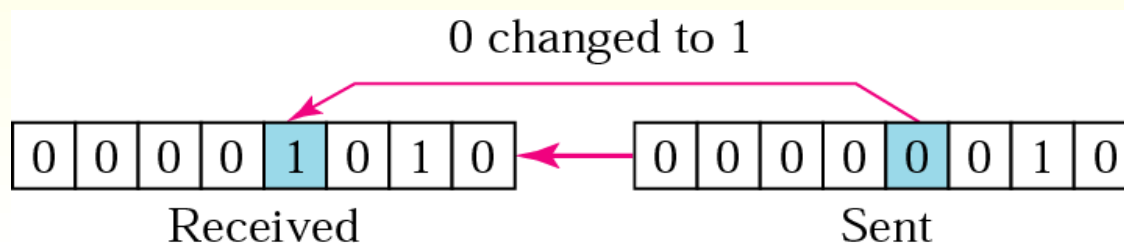
■ Introduction

- the data link layer is responsible for carrying a packet from one hop (computer or router) to the next hop; i.e., it has local responsibility unlike the network layer
- position of the data link layer

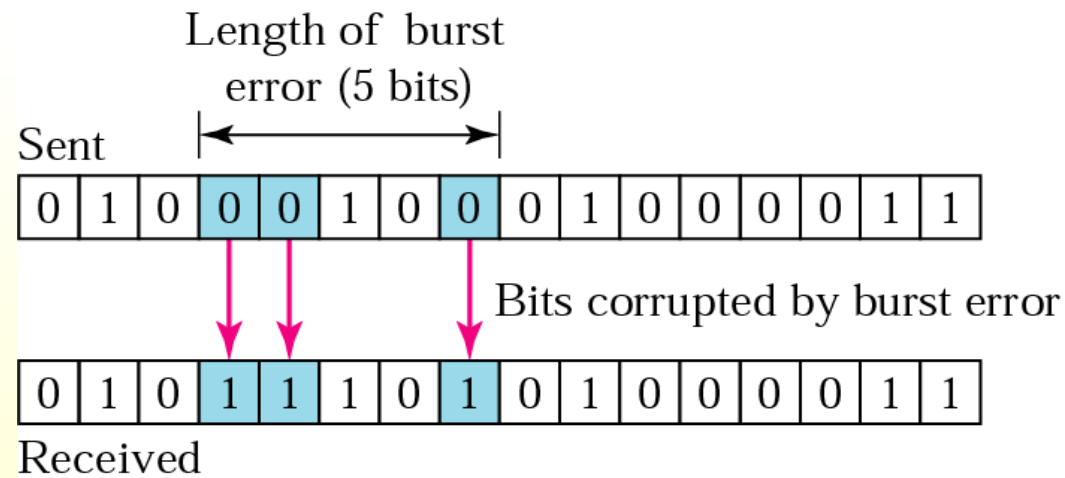


3.1 Error Detection and Correction

- mainly a data link layer function
- networks must be able to transfer data from one device to another with complete accuracy - our wish
- data can be corrupted during transmission - many factors exist - like transmission impairments
- hence, reliable systems must have a mechanism for detecting and correcting errors
- two types of errors: **single-bit** and **burst**
 - **single-bit error**: only one bit in a **data unit** (byte, character, packet, ...) has changed; less likely to occur in serial transmission (since noise normally lasts longer than the transmission time of 1 bit, e.g. 1 microsecond if speed is 1 Mbps), but most likely to occur in parallel transmission



- **burst error**: 2 or more bits in the data unit have changed; most likely to occur in serial transmissions

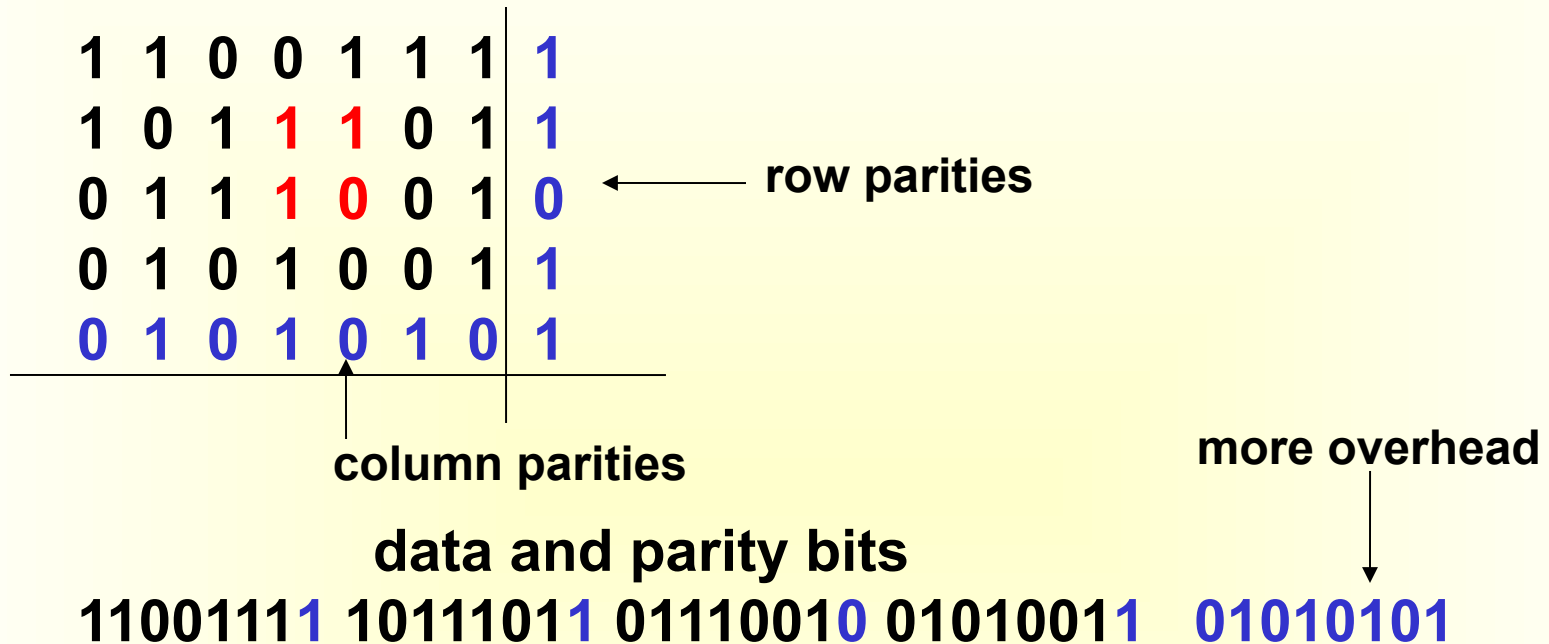


- **error detection and correction**: include **redundant** error-detection (and correction) codes with the data
 - two possibilities
 - include enough redundant information (called **error-correcting codes**) to enable the receiver to deduce what the transmitted data must have been; this method is often referred to as **forward error correction**; applicable on unreliable channels such as wireless links
 - include only enough redundant information (called **error-detecting codes**) to allow the receiver to deduce that an error occurred, but not which error, and have it request retransmission; applicable on reliable channels such as fiber

- error-detecting codes: two types: **parity check** and **cyclic redundancy check** (a third one, called **checksum**, also exists)
- **parity check**: most common and simple; two varieties
 - **simple parity check**: a redundant bit, called a **parity bit**, is added to every data unit so that the total number of 1s in the unit (including the parity bit) becomes even (or odd for odd-parity)
 - this method can detect all single-bit errors; it can detect burst errors only if the total number of errors in each data unit is odd
 - **two-dimensional parity check**: a block of bits is organized in a table (rows and columns)

original data

1100111 1011101 0111001 0101001



- can detect most errors
- if 2 bits in one data unit are damaged and two bits in exactly the same positions in another data unit are also damaged, the checker will not detect an error

- **cyclic redundancy check (CRC)**: is based on binary division
 - the CRC is appended to the end of a data unit so that the resulting data unit becomes exactly divisible by a second, predetermined binary number
 - **read about the method of generating the CRC**
- **error correction**: two methods
 - **error correction by retransmission**: when an error is detected, the receiver will tell the sender to retransmit the entire data unit; see next: **Data Link Control**
 - **forward error correction**: the receiver can use an error-correcting code to correct certain errors; the **hamming code** is used to detect and correct errors; 😊

3.2 Data Link Control and Protocols

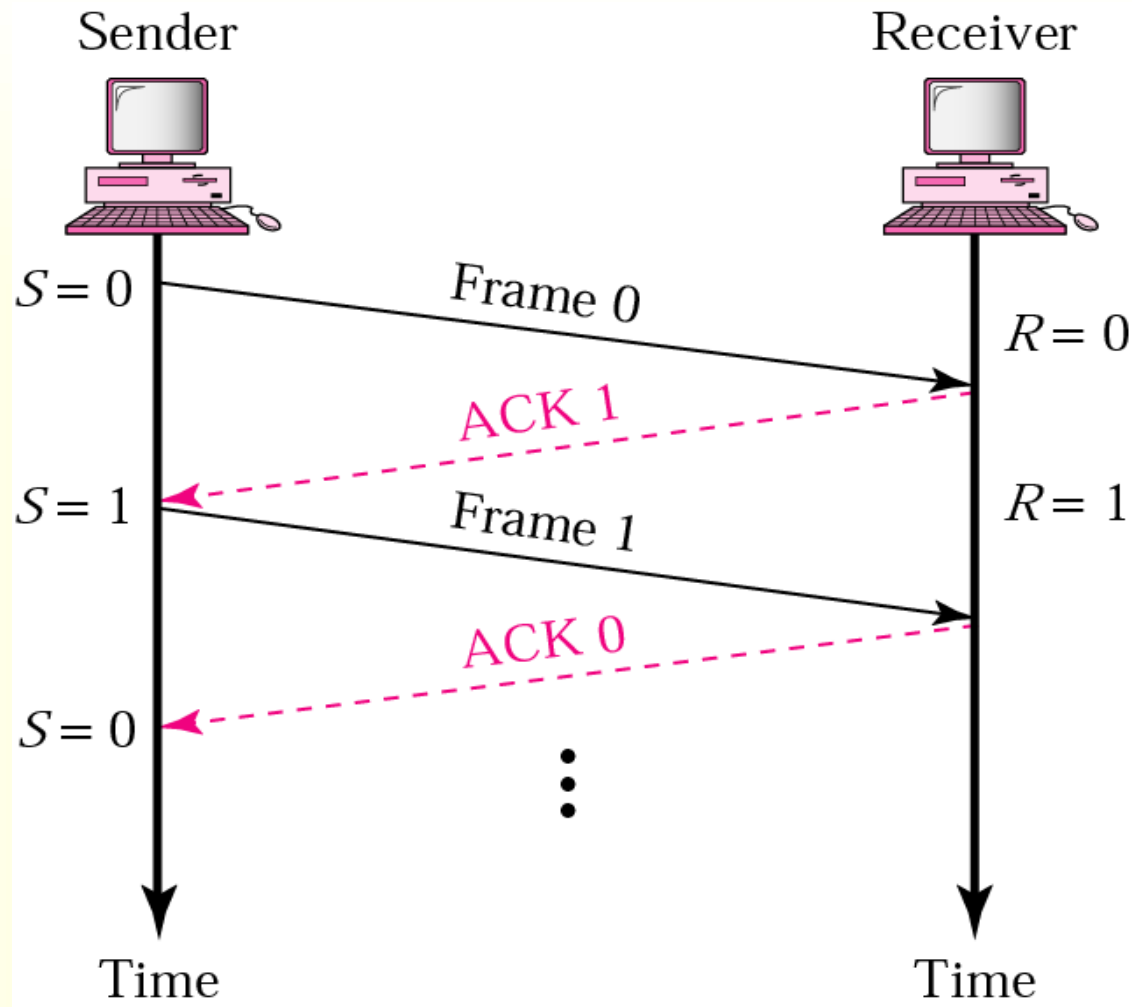
- for reliable data delivery across the link, the data link layer has two most important responsibilities: **flow control** and **error control**, collectively known as **data link control**
- **Flow Control**
 - refers to a set of procedures used to restrict the amount of data that the sender can send before receiving an acknowledgement
 - it is an end-to-end mechanism for regulating traffic between source and destination
 - **congestion control**: a mechanism used by the network to limit congestion
 - flow control and congestion control are not really separable, and we will refer to both as **flow control**

- limiting factors: **speed** of the receiver (to process incoming data), amount of **memory (buffer)** to store the incoming data before processed, and **network capacity**
 - rate of processing is normally slower than rate of transmission
 - ➔ the receiver must be able to inform the sender to temporarily halt transmission until it is ready to receive more
- if no flow control
 - when overload occurs
 - queues build up
 - packets are discarded
 - sources retransmit messages
 - congestion increases ➔ instability
- flow control prevents network instability by keeping packets waiting outside the network rather than in queues inside the network
 - avoids wasting network resources

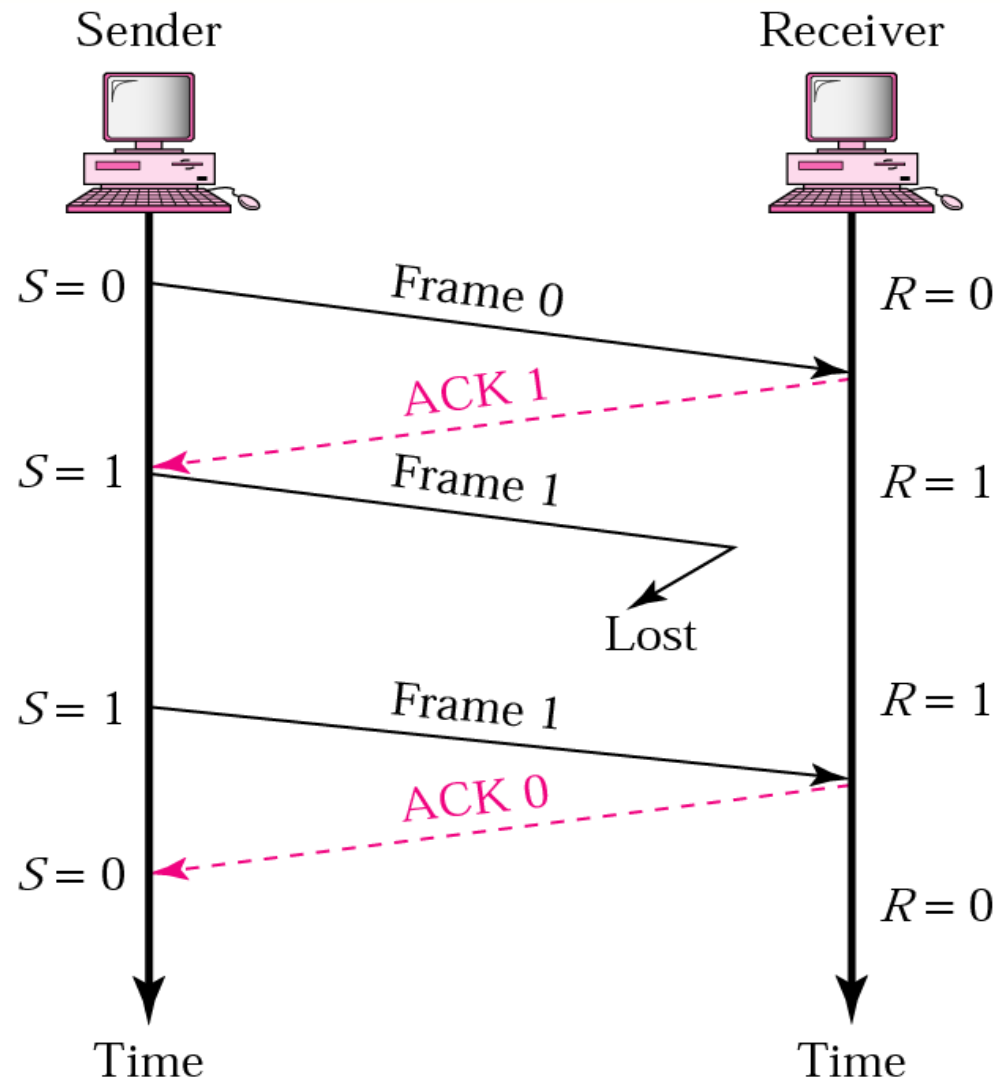
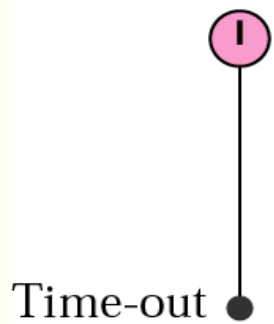
- **Error Control**
 - refers to both error detection and error correction
 - in the data link layer, error control refers primarily to methods of error detection and retransmission
 - anytime an error is detected, specified frames are retransmitted; this process is called **automatic repeat request (ARQ)**
- **Flow and Error Control Mechanisms** (three methods)
 1. **An Unrestricted Simplex Protocol**
 - simplex transmission: from sender to receiver
 - processing time ignored; the receiver is assumed to process the received data infinitely quickly
 - infinite buffer space assumed to be available
 - communication channel assumed to be free of error

2. Stop-And-Wait ARQ

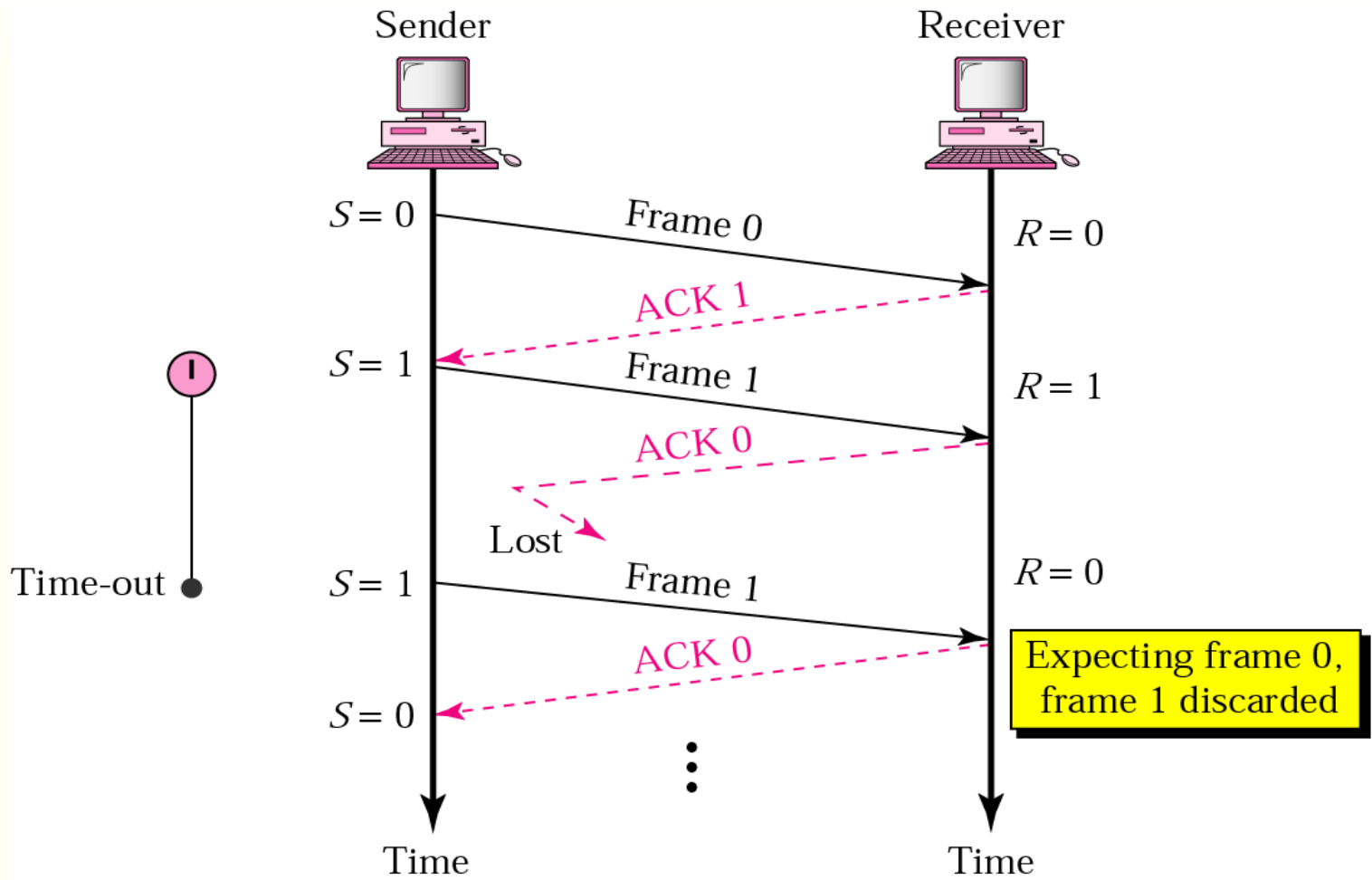
- protocols in which the sender sends one frame and then waits for an acknowledgement before proceeding are called **stop-and-wait**
- data traffic is simplex, but frames travel in both directions; hence a half-duplex physical channel suffices
- this protocol fails if ack frames by the receiver do not arrive or are late; the sender may time-out and send a frame again creating duplicate frames; the solution is to give sequence numbers to data frames and acks
- there are four situations: normal operation, the frame is lost, the ack is lost, the ack is delayed



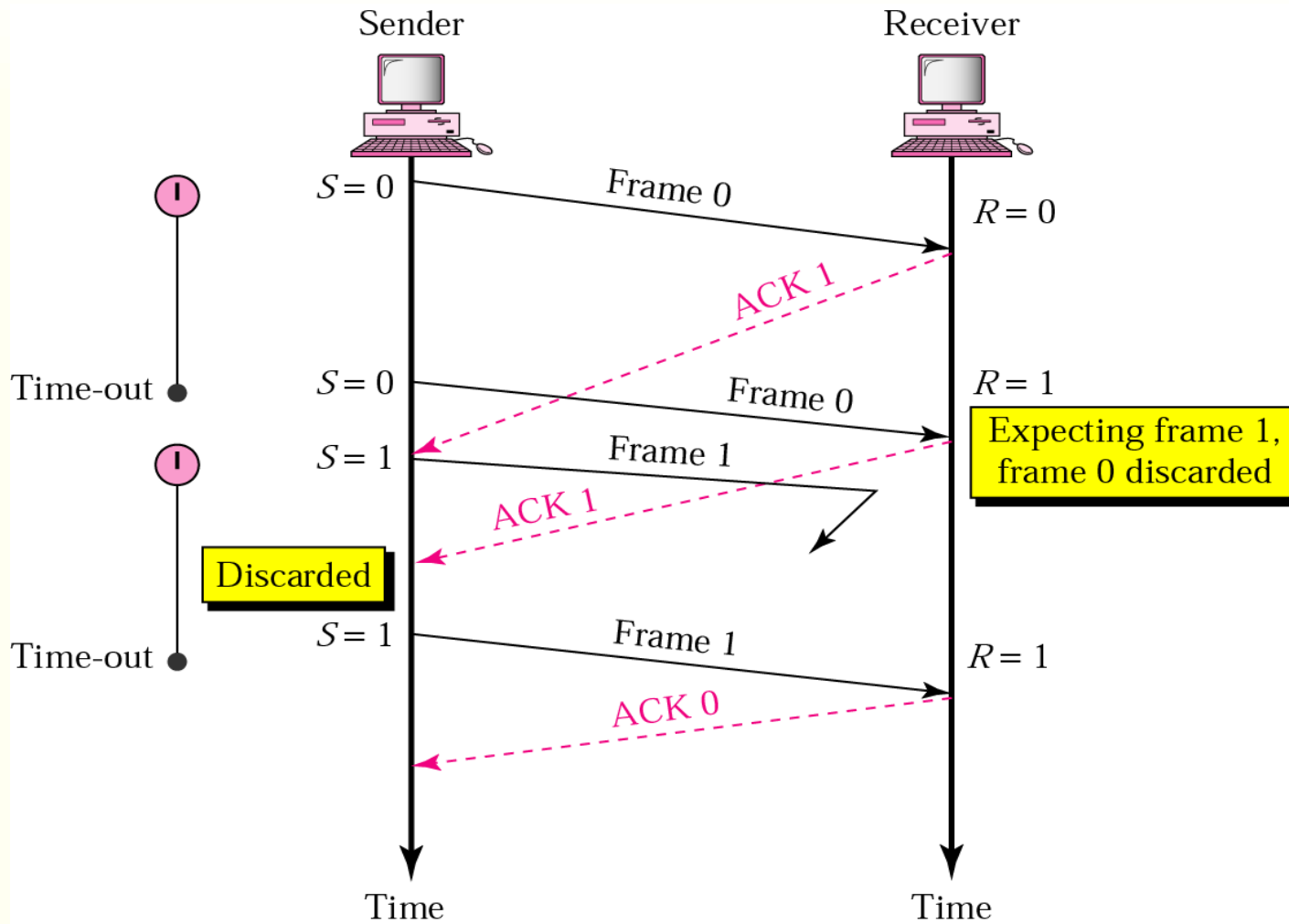
***Normal
operation***



Lost frame



Lost ack



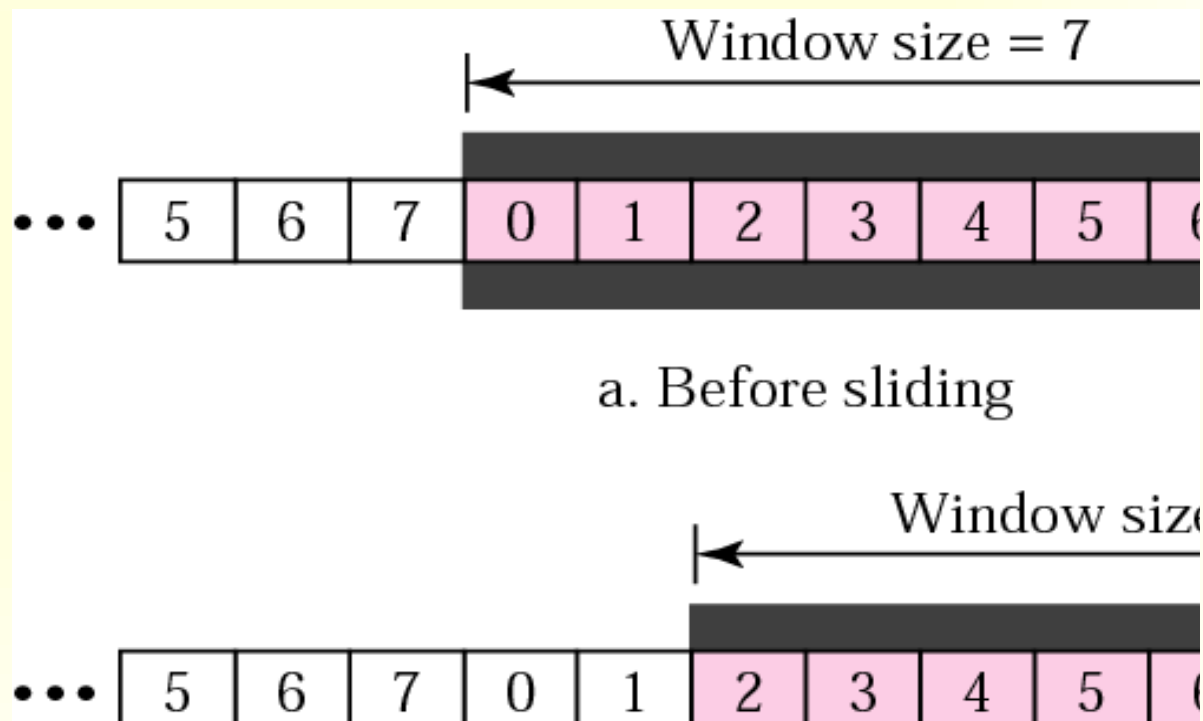
Delayed ack

3. Sliding Window Protocols

- Stop-and-wait ARQ is not efficient; there is only one frame that is sent
- can we transmit multiple frames while waiting for an ack?
- use full-duplex transmission
- two protocols: **Go-Back-N ARQ** and **Selective Repeat ARQ**
- both methods require frames to be numbered
- **sequence numbers**
 - frames from a sending station are numbered sequentially from 0 to 2^m-1 , where **m** is the number of bits of the sequence number in the frame header
 - e.g., let $m = 3$, then frames can be numbered as
0, 1, 2, 3, 4, 5, 6, 7, 0, 1, 2, 3, 4, 5, 6, 7, 0, 1, ...

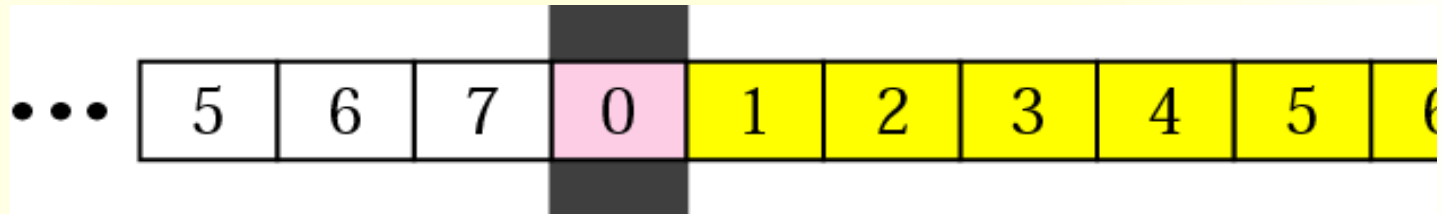
a. **Go-Back-N-ARQ**

- the sender and receiver have sliding windows
- the sender's window is of fixed size and is at most 2^m-1 ; but other protocols such as TCP allow variable size windows (see later in the transport layer); it holds frames sent but not acknowledged
- the receiver's window is of size 1
- the sender sets a timer for each frame sent; the receiver has no timer
- the receiver sends positive acknowledgements if a frame has arrived safe and in order
- if a frame is damaged or out of order, the receiver discards it and keeps silent and discards all subsequent frames; this causes the timer of the sender to expire and **goes back** and resends all frames, beginning from the one with the expired timer
- the receiver does not need to acknowledge frames individually



Sender sliding window

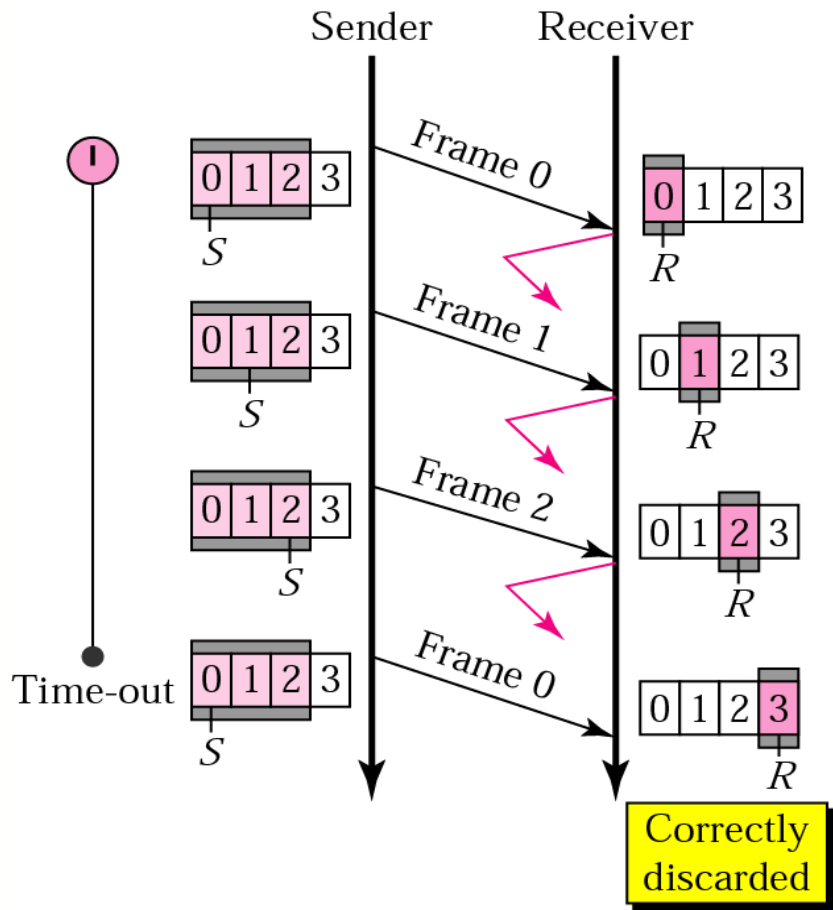
waiting for
frame 0



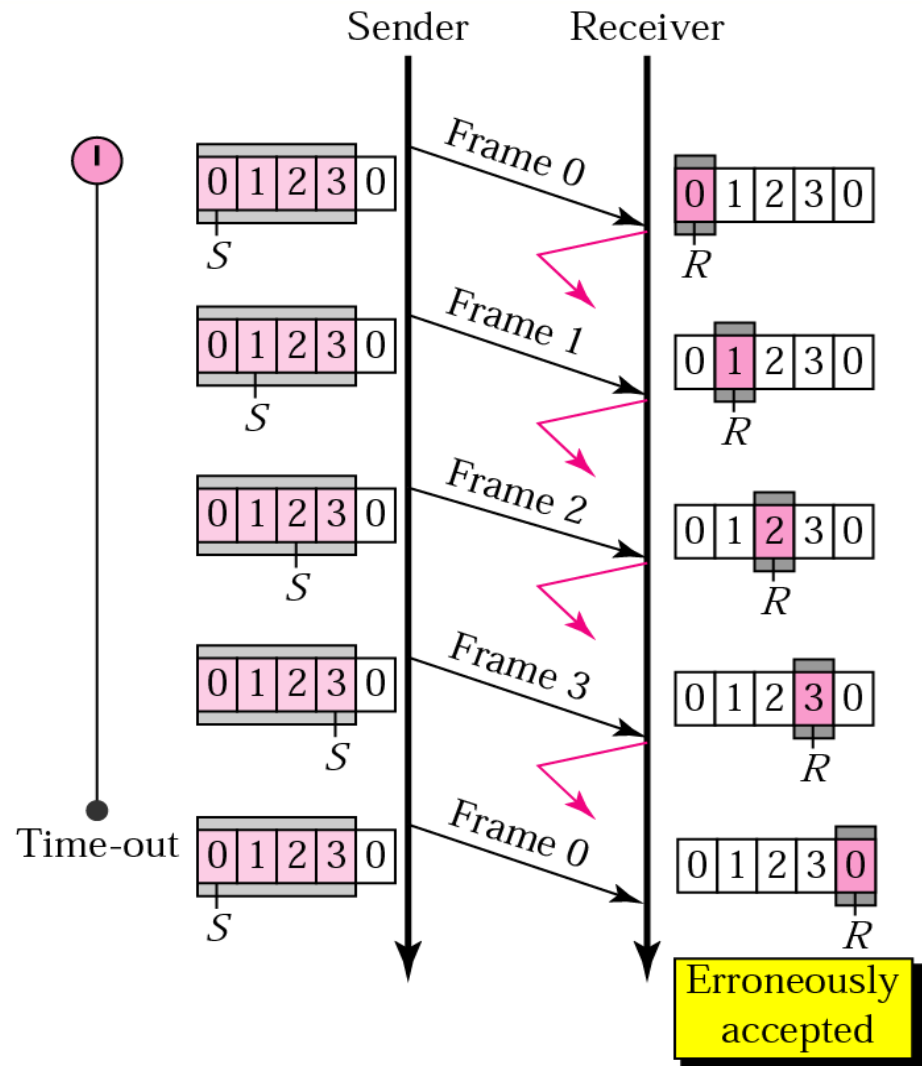
a. Before sliding

Receiver sliding window

- **Why is the sender's window $< 2^m$?**
 - **To detect repeated frames**
- **Let $m = 2$**
- a. **If the size of the window is 3 ($2^m - 1$) and all the 3 acks are lost, the timer for frame 0 expires and all frames are resent; the window of the receiver is expecting frame 3 not frame 0; so it will discard the duplicated frame**
- b. **If the size of the window is 4 (2^m) and all the 4 acks are lost, the sender will send the duplicate of frame 0; since the window of the receiver is expecting frame 0, it accepts it not as a duplicate but as the first frame in the next cycle**



a. Window size $< 2^m$

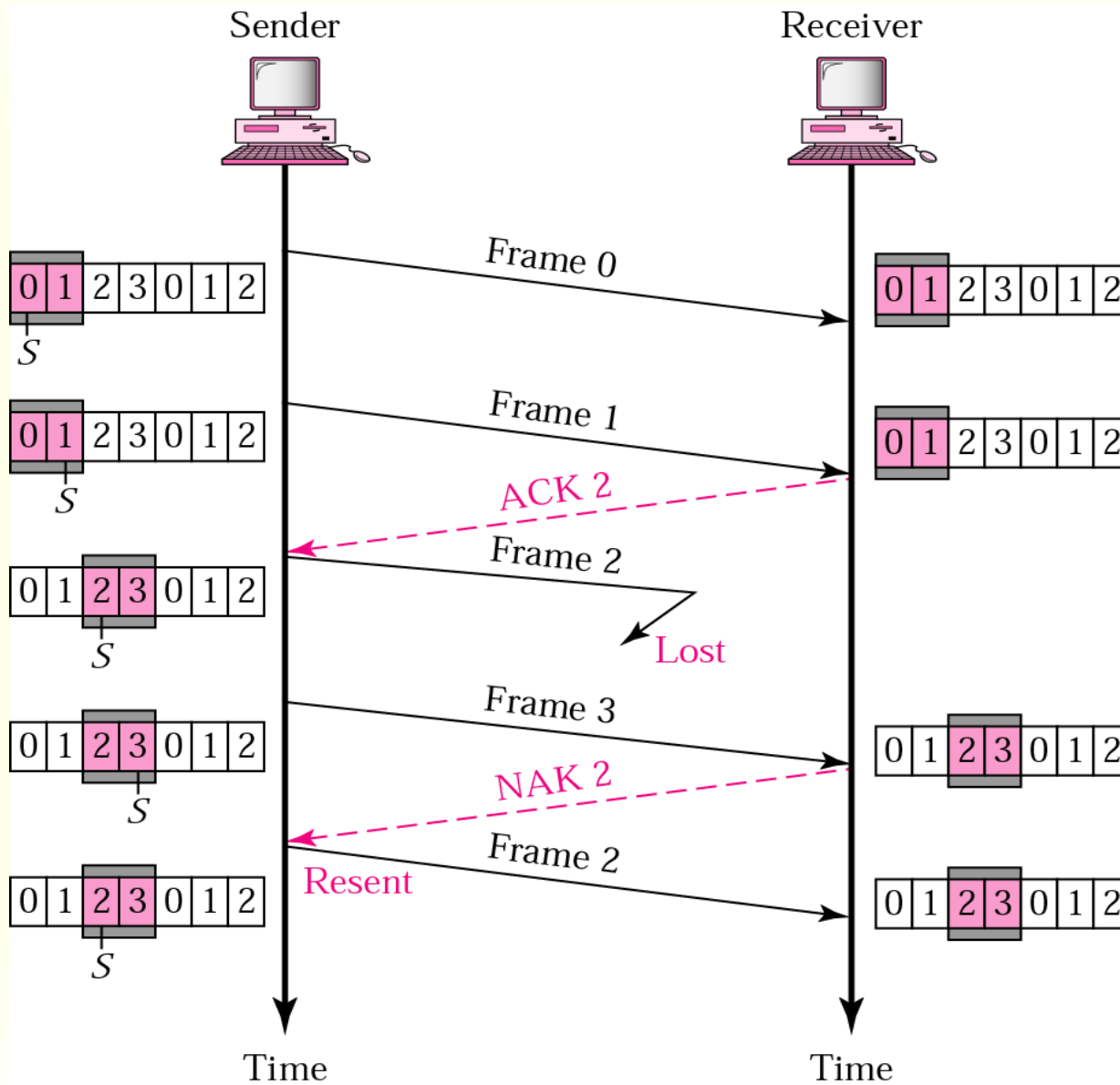


b. Window size $= 2^m$

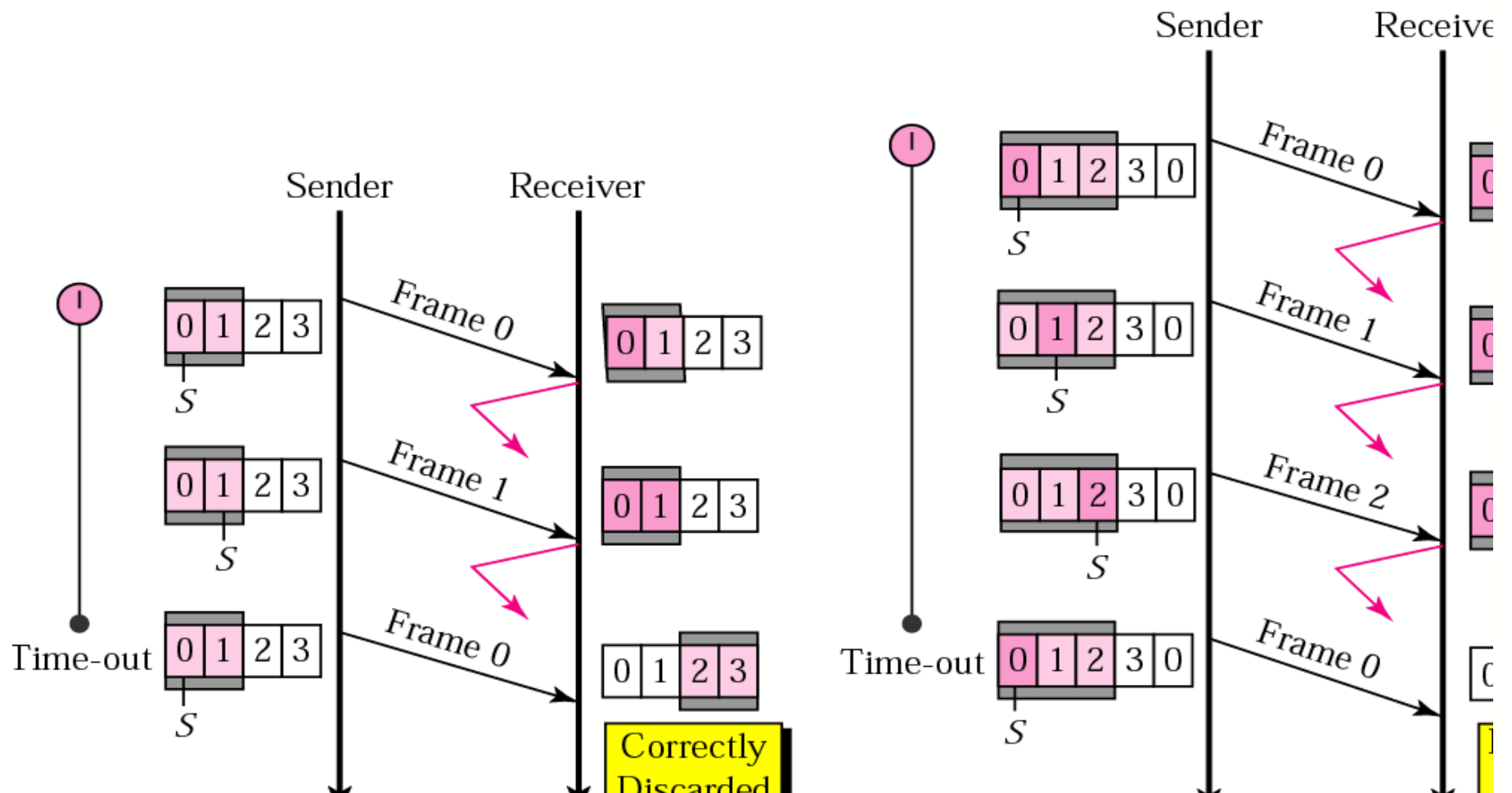
b. Selective Repeat ARQ

- in GO-Back- N ARQ, the task of the receiver is simplified; no need to buffer out-of-order frames; they are simply discarded
- this protocol is inefficient for a noisy link that has a high probability of damage resulting in the resending of multiple frames
- for a noisy link, use Selective Repeat ARQ that requires resending only the damaged frame
- Selective Repeat ARQ defines a negative ack (NAK) that reports the sequence number of a damaged frame before the timer expires
- the sender and receiver windows are both at most half of 2^m
- the receiver's window defines the range of acceptable sequence numbers, i.e., frames may arrive out of order

- Let $m = 2$; window sizes = 2



- **Why are the sender and receiver windows at most half of 2^m ?**
 - **To detect repeated frames**
- **Let $m = 2$**
- a. **If the size of the window is 2 and both acks are lost, the timer for frame 0 expires and frame 0 is sent; the window of the receiver is expecting frame 2 not frame 0; so it will discard the duplicated frame**
- b. **If the size of the window is 3 and all the 3 acks are lost, the sender sends a duplicate of frame 0; since the window of the receiver is expecting frame 0 (frame 0 is part of the window), it accepts it not as a duplicate but as the first frame in the next cycle**



- **examples of data link protocols**
 - **HDLC** - High-level Data Link Control
 - an ISO standard
 - an actual protocol designed to support both half-duplex and full-duplex communication over point-to-point and multipoint links that implements the ARQ mechanisms
 - **PPP** - Point-to-Point Protocol
 - used in the Internet for router-to-router and home user-to-ISP traffic

3.3 Medium Access Control

- The **MAC Sublayer**: in a network, two devices can be connected by a dedicated link (like a reserved highway, may be on certain occasions!) or a shared link (like a public highway)
 - **point-to-point access**: when two devices are connected by a dedicated link and this link can be used by them at any time; PPP is used
 - **multiple access**: when two devices are connected by a shared link; when two devices in a multiple access situation get access to the link or a **channel** in the link, they may need to use a point-to-point access protocol to exchange data
- **Multiple Access Protocols**
 - the problem of controlling the access to the medium is similar to the rules of speaking in an assembly (the right to speak is upheld; two people do not speak at the same time; do not interrupt each other; do not monopolize the discussion; ...)

- **three categories of multiple access protocols**
 - 1. Random Access Protocols - try your best like taxis do**
 - i. MA - Multiple Access**
 - ii. CSMA - Carrier Sense MA**
 - iii. CSMA/CD - CSMA with Collision Detection**
 - iv. CSMA/CA - CSMA with Collision Avoidance**
 - 2. Controlled-Access Protocols - get permission**
 - i. Reservation**
 - ii. Polling**
 - iii. Token Passing**
 - 3. Channelization Protocols - simultaneous use**
 - i. FDMA - Frequency-Division MA**
 - ii. TDMA - Time-Division MA**
 - iii. CDMA - Code-Division MA**

1. Random Access Protocols

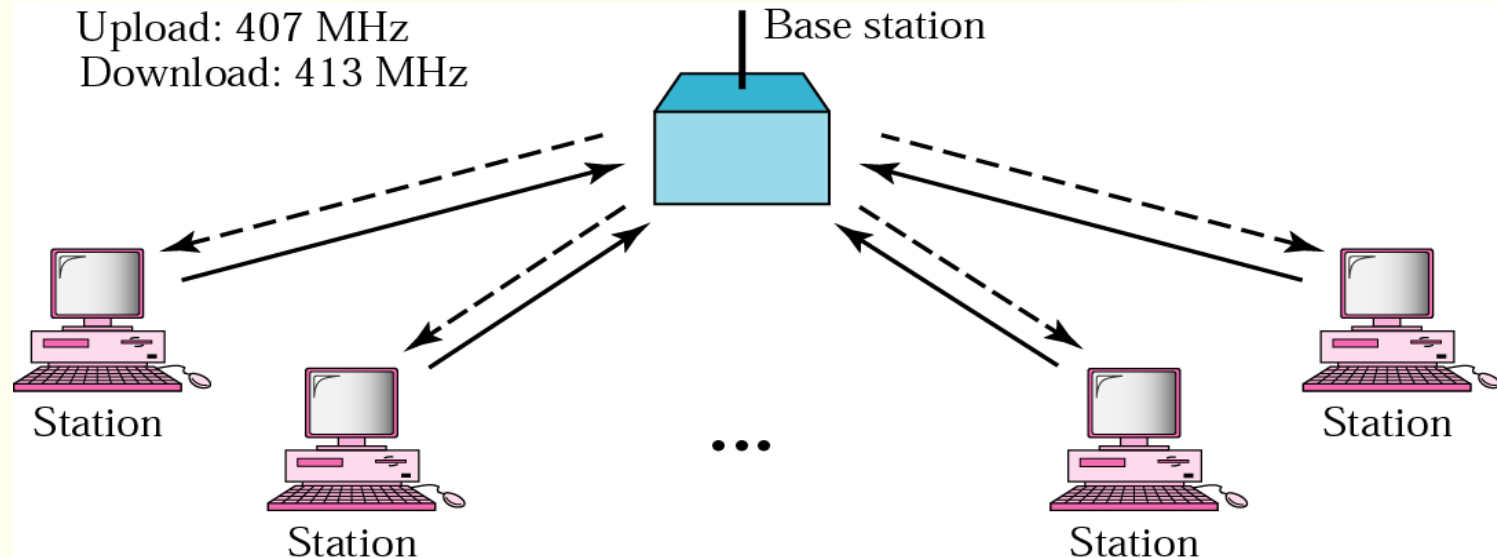
- each station has the right to use the medium without being controlled by any other station
- **collision** may occur if more than one station tries to send
- we need a procedure to answer the following questions
 - when can a station access the medium?
 - what can the station do if the medium is busy?
 - how can the station determine the success or failure of the transmission?
 - what can the station do if there is an access conflict?

i. MA - Multiple Access

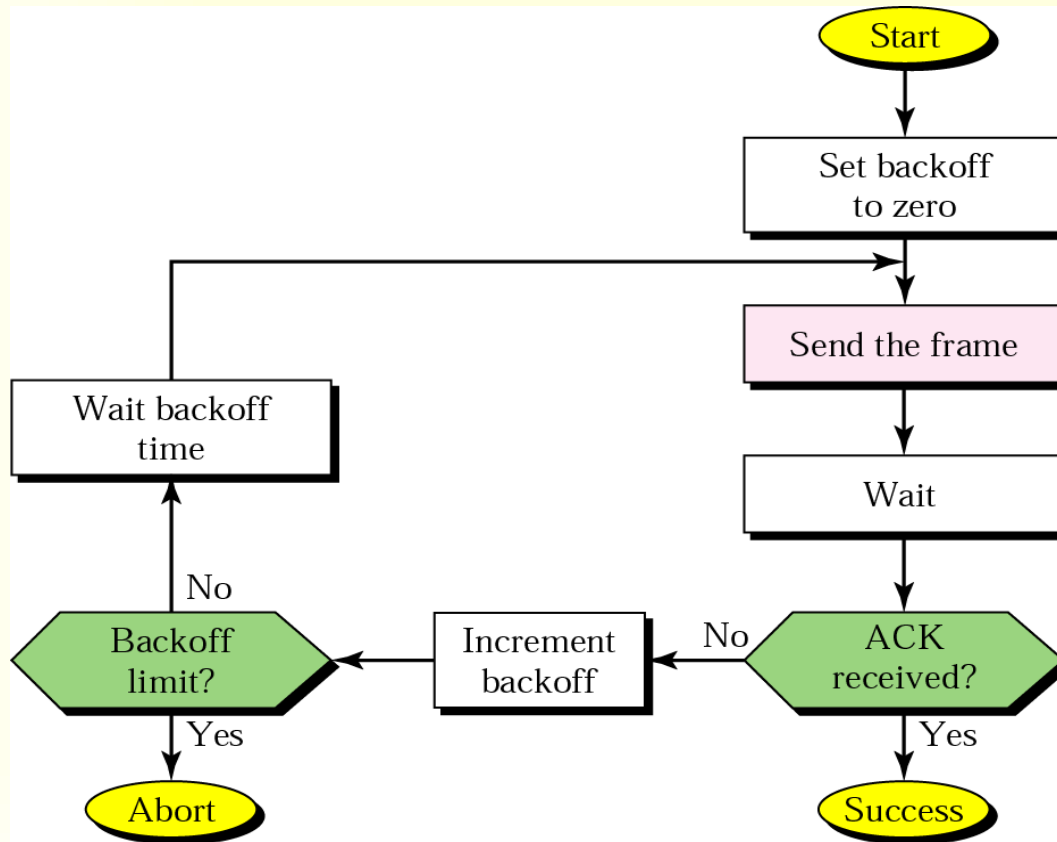
- ALOHA is the earliest RA method developed at the University of Hawaii in the early 1970s
- originally designed to be used on a radio LAN with a data rate of 9600 bps
- can also be used in satellite and wireless transmissions

*Aloha (in Hawaiian language) means **hello** and **goodbye***

- a base station serves as a central controller; each station sends a frame to the base station with an uploading frequency of 407 MHz (the solid line in the figure); the base station sends it to the receiver using a downloading frequency of 413 MHz (the dashed line in the figure);



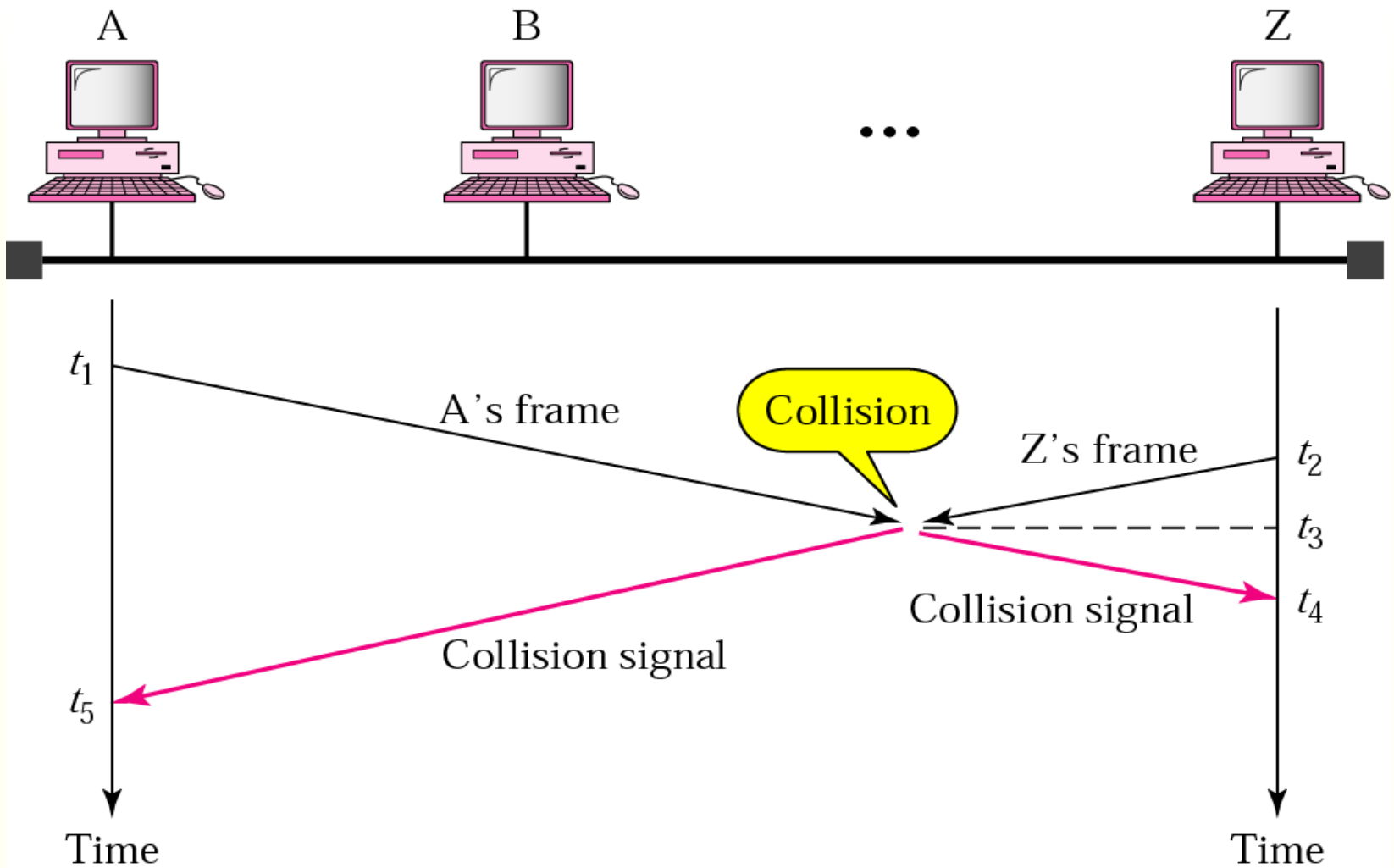
- potential collisions exist (on the upload link); so wait for acknowledgement; if none arrives wait a period of time (2 times the maximum propagation delay), send again if **back off** limit did not reach; the time it waits increases from one trial to the next



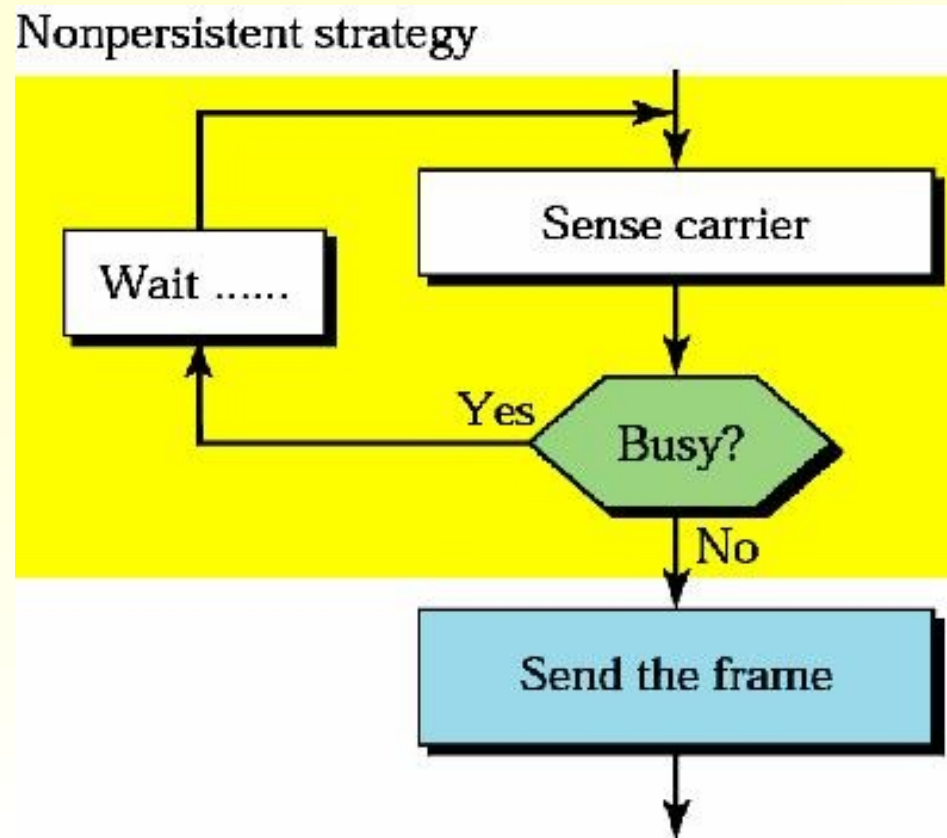
- **versions**
 - **pure ALOHA**
 - **slotted ALOHA**
 - **time is divided into discrete intervals of one packet duration**
 - **transmitting a frame is allowed only at the beginning of a slot**
 - **still collision is possible; collided packets are retransmitted after a random delay**

ii. CSMA - Carrier Sense MA - polite version of ALOHA

- **to minimize the chance of collision, each station first listens to the medium before sending; “listen before talk”**
- **if the channel is busy, it waits until it is idle**
- **otherwise it transmits; if a collision occurs, it waits a random amount of time and starts listening again**
- **the chance of collision is minimized**
 - **but may still occur because of the propagation delay (a station doesn't know if another one has just started transmitting) or**
 - **if two or more stations start transmitting at the same time**

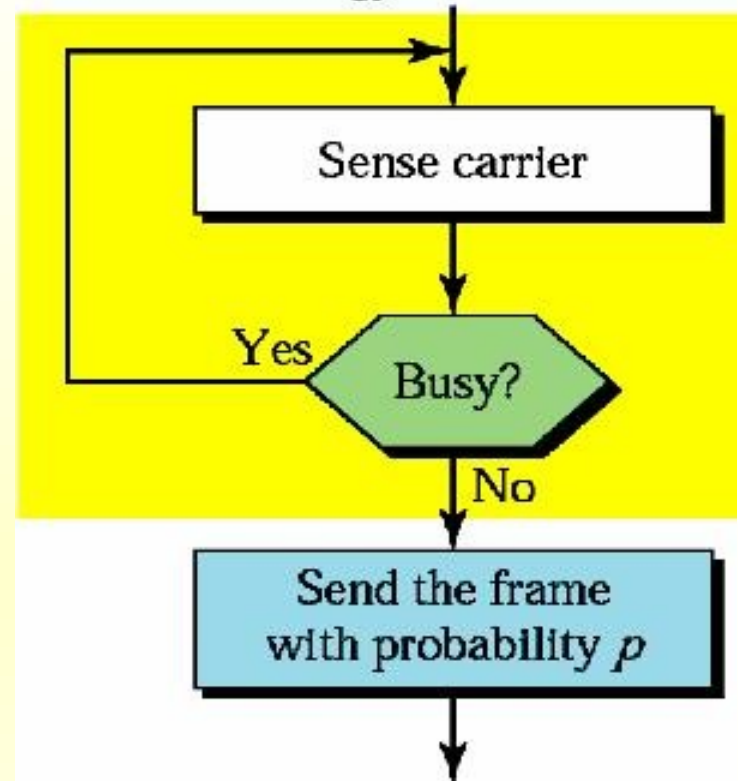


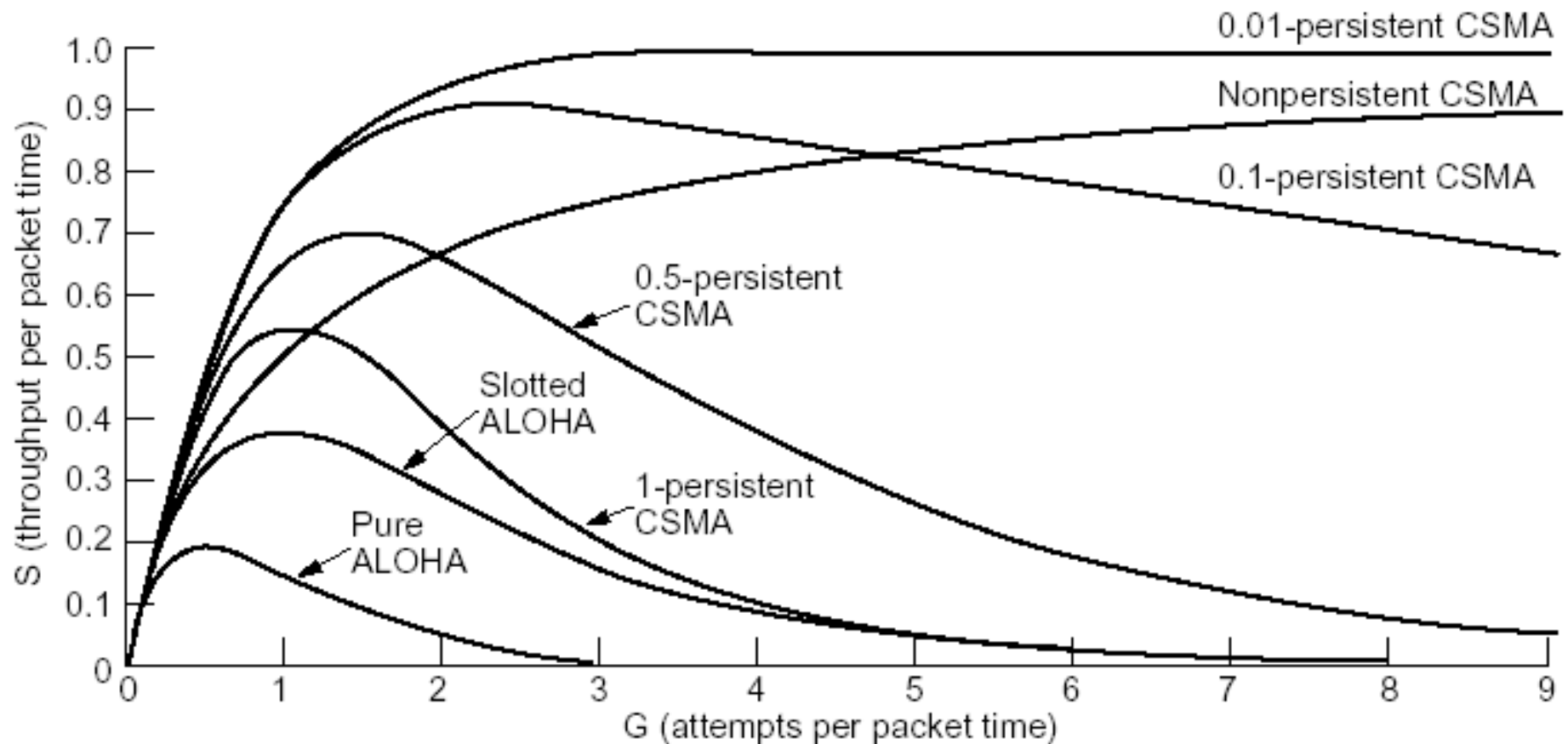
- persistence strategies: two substrategies have been defined
 - **nonpersistent**: sense a line and send if it is idle; otherwise wait a **random** amount of time before listening (hence less **greedy** than continuously listening); reduces the chance of collision, but also reduces the efficiency of the network and has longer delays



- **persistent**: sense a line and “send” if it is idle; otherwise listen; two variations of sending
 - **1-persistent**: if the line is idle, send **immediately** (with probability 1)
 - **p-persistent**: if the line is idle, send with probability p and refrain from sending with probability $1-p$; if $p = 0.2$, then a station sends 20% of the time that a line is idle, and refrains from sending 80% of the time
 - implementation: the station generates a random number between 1 and 100; if it is < 20 , then it sends, otherwise it refrains
- it reduces the chance of collision and improves efficiency; depends on the value of p (see next figure)

Persistent strategy

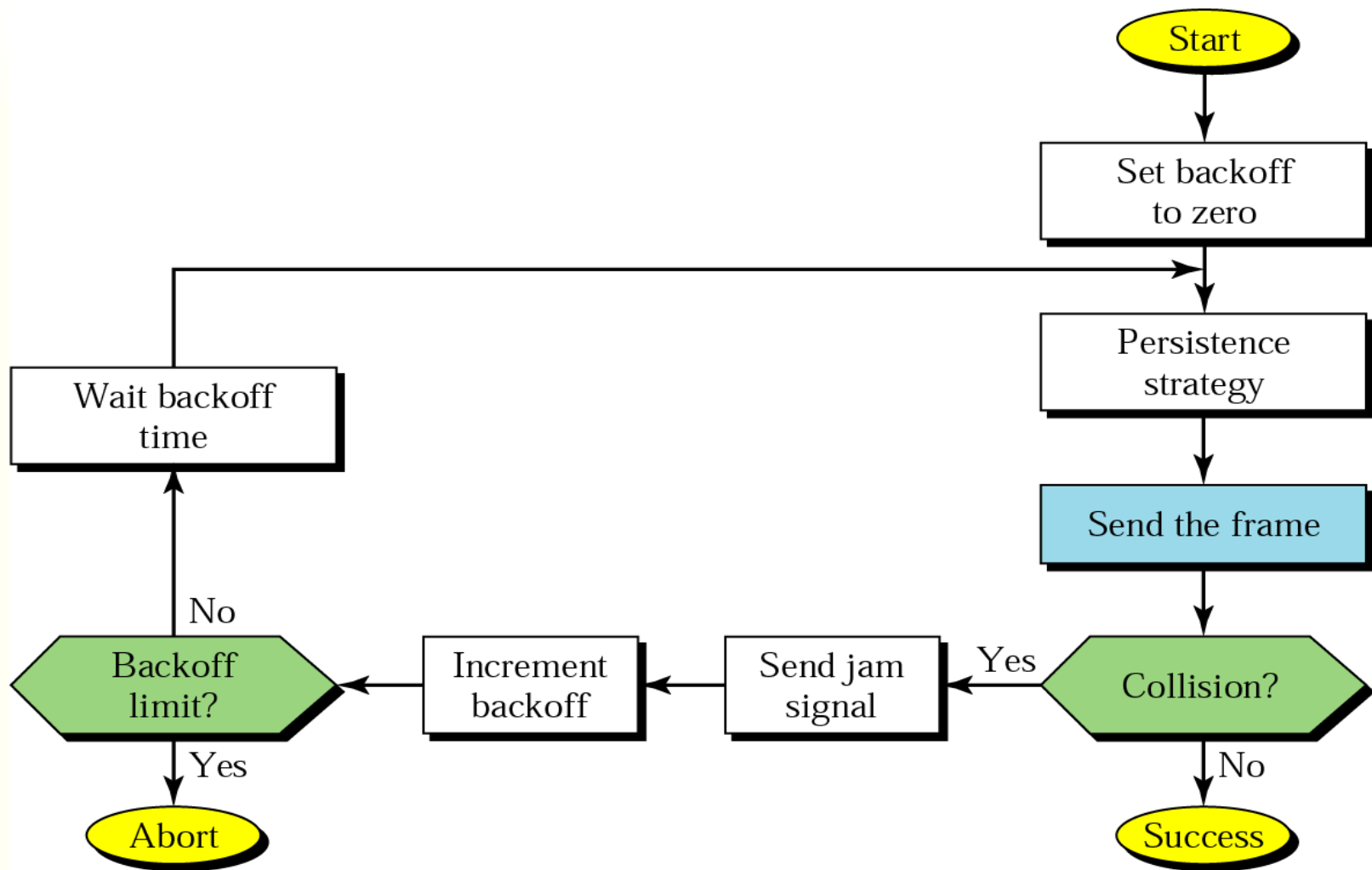




computed throughput versus offered traffic for the protocols discussed so far

iii. CSMA/CD - CSMA with Collision Detection

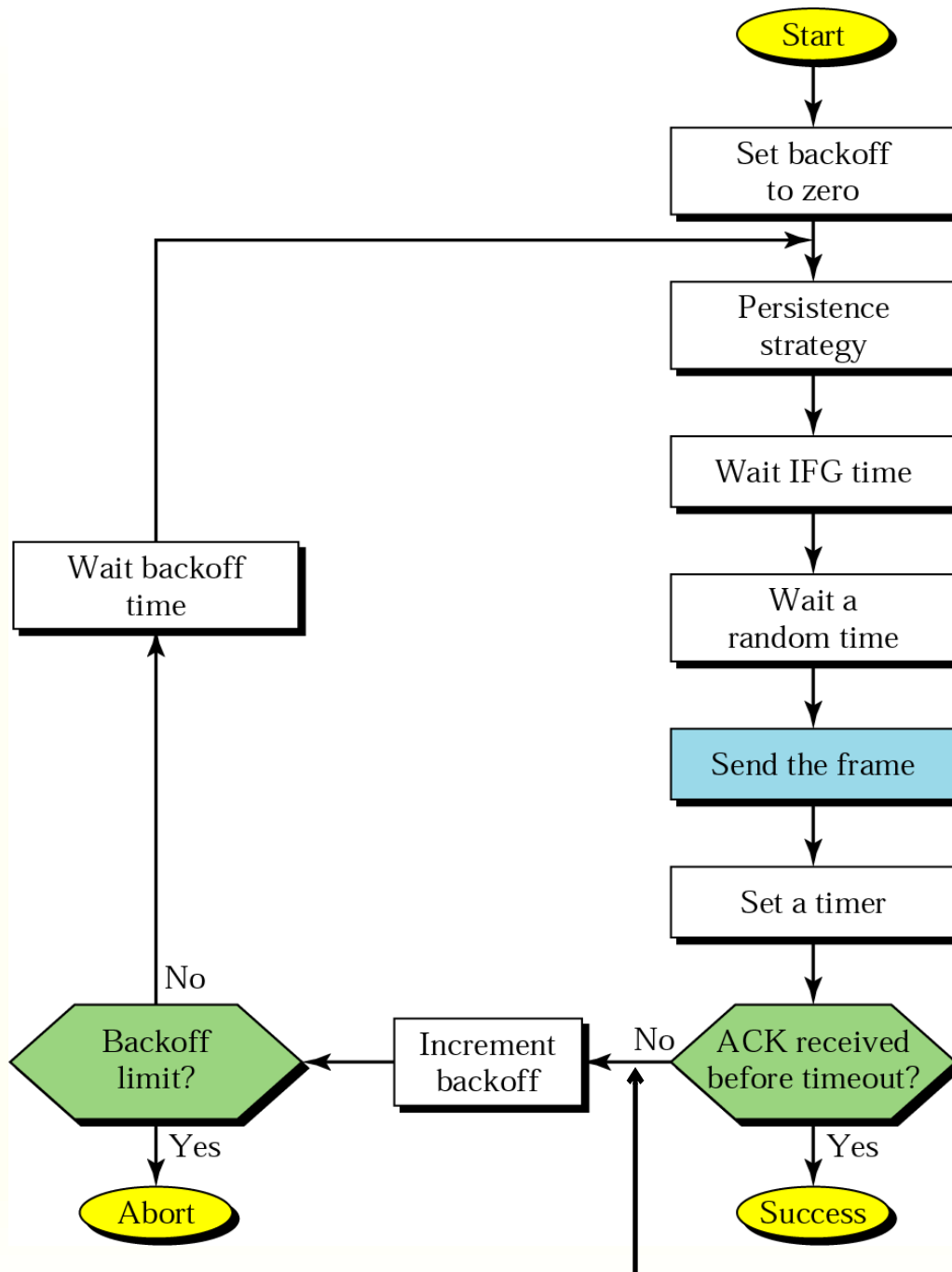
- adds a procedure to handle a collision
- if a collision is detected and to reduce the probability of collision the second time, the sender waits; it has to **back off**
- it waits a little the first time, more if a collision occurs again, much more if it happens a third time, and so on; finally gives up
- in the exponential backoff method, it waits an amount of time between 0 and $2^N \times \text{maximum_propagation_time}$, where N is the number of attempted transmissions
- line sensing is done using one of the persistent strategies



- sending a jam alerts the other stations and also to discard the part of the frame received
- used in traditional Ethernet; CSMA was never implemented

iv. CSMA/CA - CSMA with Collision Avoidance

- **avoids collision**
- **uses one of the persistence strategies; after it finds the line idle, it waits an IFG (interframe gap) amount of time; it then waits another random amount of time; after that it sends the frame and sets a timer;**
- **if it receives an ack before the timer expires, the transmission is successful; otherwise something is wrong (the frame or the ack is lost); waits for a backoff amount of time and re-senses the line**
- **used in wireless LANs**



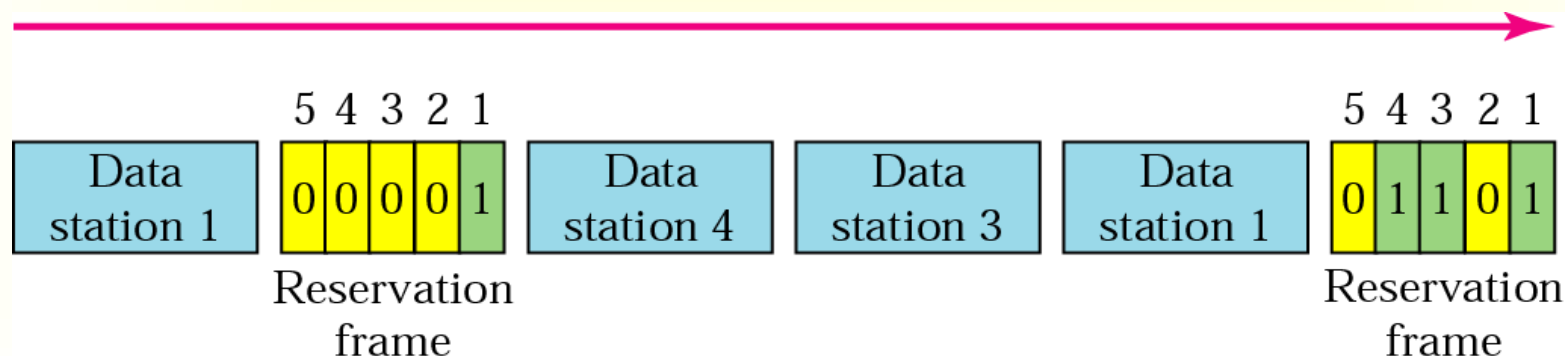
■ the frame or the ack is

2. Controlled-Access Protocols

- the stations consult one another to find which station has the right to send; a station can not send unless it has been authorized by other stations

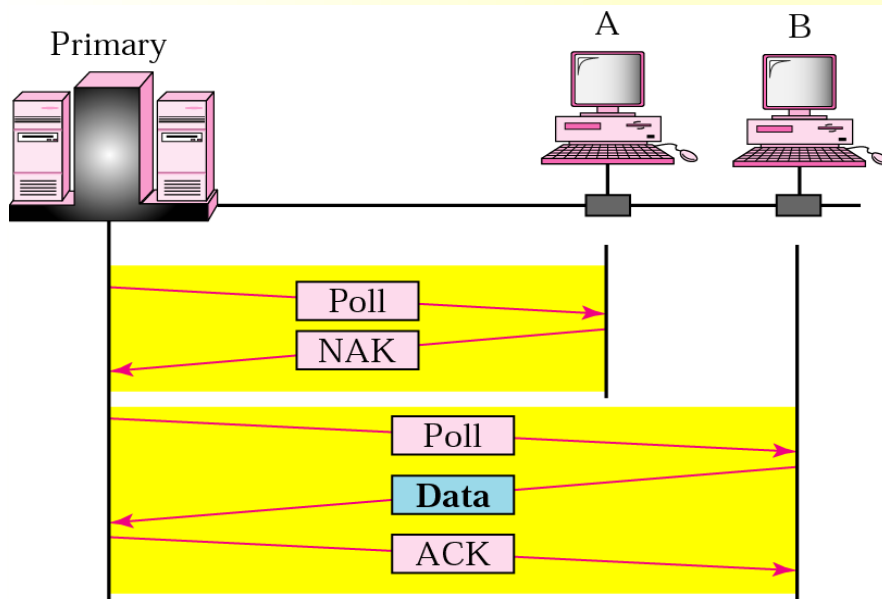
i. Reservation

- a station needs to make a reservation before sending data
- time is divided into intervals; in each interval, a reservation frame precedes the data frames sent in that interval; if there are N stations in the system, there are exactly N reservation minislots in the reservation frame

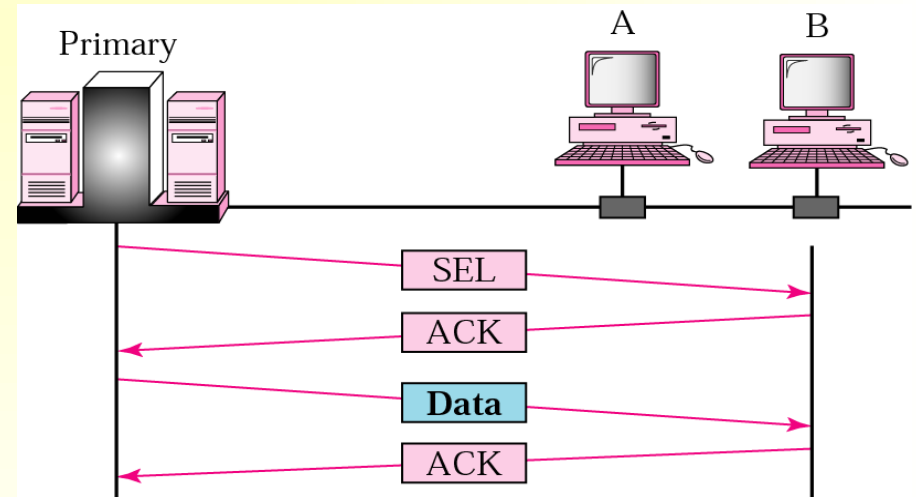


ii. Polling

- for topologies in which one device is designated as a **primary station** and the other stations are **secondary stations**
- the primary asks the secondaries if they have data to send (**polling**); when it has data to be sent, the primary tells the secondary to get ready to receive (**selecting**)



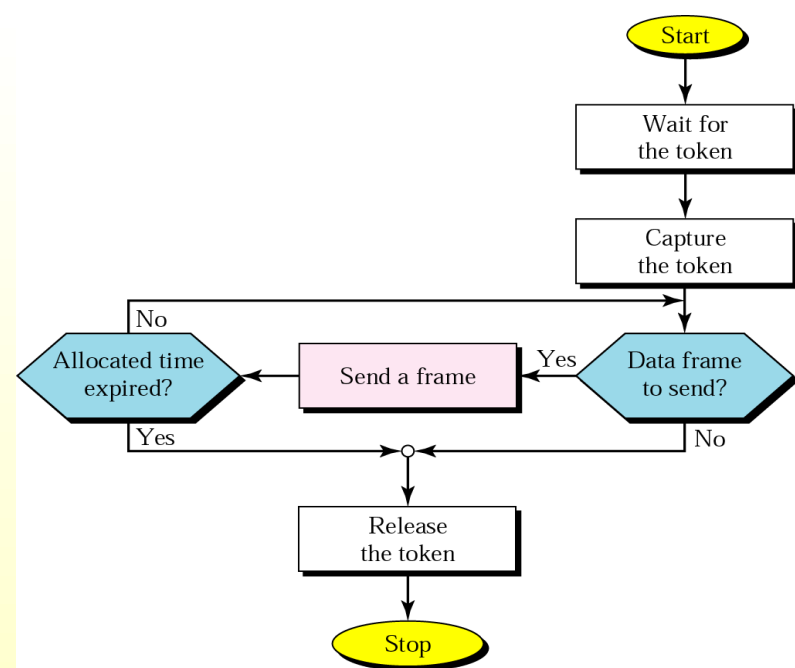
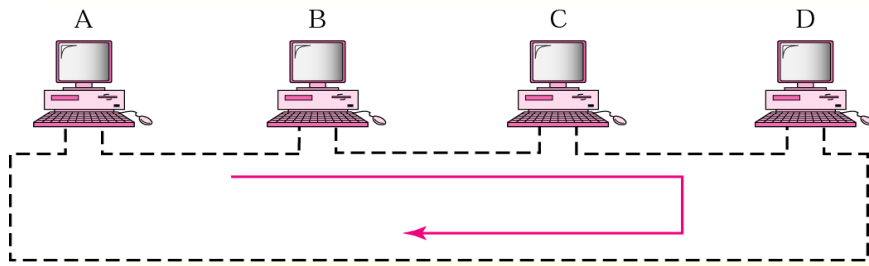
a. poll



b. select

iii. Token Passing

- a station is authorized to send data when it receives a special frame called a **token**
- the stations are arranged around a ring (each station has a predecessor and a successor)
- a token circulates around the ring when no data is transmitted
- token: a bit sequence
 - free token: 01111110
 - busy token: 01111111
- when a node wants to transmit
 - wait for a free token
 - remove token from ring (replace with busy token)
 - transmit message
 - when done transmitting, replace free token on ring



- **token failures:** tokens can be created or destroyed by noise
- **distributed solution**
 - nodes are allowed to recognize the loss of a token and create a new token
 - collision occurs when two or more nodes create a new token at the same time → need collision resolution algorithms
- **node failures:** since each node must relay all incoming data, the failure of a single node will disrupt the operation of the ring

- **example**
 - **FDDI - Fiber Distributed Data Interface** - is a 100 Mbps fiber optic token ring LAN standard - see later in LANs
 - it is also used for MANs

3. Channelization Protocols

- the available bandwidth of a link is shared in **time**, **frequency**, or through **code**, between different stations

i. FDMA - Frequency-Division MA

- the available bandwidth is divided into channels; each station uses its allocated band to send its data; each band is reserved for a specific station (it belongs to it all the time)
- FDMA is a data link layer protocol that uses FDM at the physical layer
- used in cellular telephone and satellite networks

iii. TDMA- Time-Division MA

- the entire bandwidth is just one channel; the stations share the capacity of the channel in time; each station is allocated a time slot during which it can send data;
- TDMA is a data link layer protocol that uses TDM at the physical layer
- also used in cellular telephone

iii. CDMA- Code-Division MA

- differs from FDMA because only one channel occupies the entire bandwidth of the link; differs from TDMA because all stations can send data simultaneously (no time sharing)
- based on coding theory; proposed several decades ago, but implemented recently due to advances in technology